

1. Create a program to store user input data into database using Scanner and JDBC.

Query:

```
create table students(  
    id int,  
    name varchar(50),  
    course varchar(50)  
);
```

Code:

```
import java.sql.*;  
import java.util.Scanner;  
public class InsertStudent {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        try {  
            Connection con = DriverManager.getConnection(  
                "jdbc:mysql://127.0.0.1:3306/practice",  
                "root",  
                "root@123");  
            System.out.println("Enter Student ID:");  
            int id = sc.nextInt();  
            sc.nextLine();  
            System.out.println("Enter Student Name:");  
            String name = sc.nextLine();  
            System.out.println("Enter Course:");  
            String course = sc.nextLine();  
            String query = "insert into students values(?,?,?)";  
            PreparedStatement ps = con.prepareStatement(query);  
            ps.setInt(1, id);
```

```

        ps.setString(2, name);
        ps.setString(3, course);
        int rows = ps.executeUpdate();
        System.out.println(rows + " row inserted");
        con.close();
    } catch (Exception e) {
        System.out.println(e);
    }
}
}

```

2. Create a program to validate login using username & password from database.

Query:

```

create table credentials(
    username varchar(50),
    password varchar(50)
);

```

```

insert into credentials values('admin','admin123');

```

Code:

```

import java.sql.*;
import java.util.Scanner;

public class LoginValidation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://127.0.0.1:3306/practice",

```

```

        "root",
        "root@123");

System.out.println("Enter Username:");

String username = sc.nextLine();

System.out.println("Enter Password:");

String password = sc.nextLine();

String query = "select * from credentials where username=? and password=?";

PreparedStatement ps = con.prepareStatement(query);

ps.setString(1, username);

ps.setString(2, password);

ResultSet rs = ps.executeQuery();

if (rs.next()) {

    System.out.println("Login Successful");

} else {

    System.out.println("Invalid Username or Password");

}

con.close();

} catch (Exception e) {

    System.out.println(e);

}

}

}

```

3. Write a program to insert multiple records using JDBC.

Query:

```

create table students(

    id int,

    name varchar(50),

    course varchar(50)

```

```
);
```

Code:

```
import java.sql.*;

public class MultipleInsert {

    public static void main(String[] args) {

        try {

            Connection con = DriverManager.getConnection(

                "jdbc:mysql://127.0.0.1:3306/practice",

                "root",

                "root@123");

            Statement st = con.createStatement();

            st.executeUpdate("insert into students values(101,'Rahul','Java')");

            st.executeUpdate("insert into students values(102,'Amit','Python')");

            st.executeUpdate("insert into students values(103,'Neha','Testing')");

            System.out.println("Multiple records inserted");

            con.close();

        } catch (Exception e) {

            System.out.println(e);

        }

    }

}
```

4. Create a program to display all records from students table.

Query:

```
create table students(

    id int,

    name varchar(50),

    course varchar(50)
```

```
);
```

Code:

```
import java.sql.*;

public class DisplayStudents {

    public static void main(String[] args) {

        try {

            Connection con = DriverManager.getConnection(

                "jdbc:mysql://127.0.0.1:3306/practice",

                "root",

                "root@123");

            Statement st = con.createStatement();

            ResultSet rs = st.executeQuery("select * from students");

            while (rs.next()) {

                System.out.println(

                    rs.getInt(1) + " "

                    + rs.getString(2) + " "

                    + rs.getString(3));

            }

            con.close();

        } catch (Exception e) {

            System.out.println(e);

        }

    }

}
```